

Turing(Meta and Google)

Work Experience

September 2024 – Present

Working on Meta's llama and Google's Gemini for their AI projects to improve the LLM performance

- Solving hard problem related to prompting and evaluating result in multi steps
- Helping Meta to improve llama performance in multi turn LLM response
- Implemented the research paper <https://arxiv.org/abs/2410.15553> issued by Meta for RLHF
- Comparing the LLM responses at different temperatures and point out major issues
- Using Sandbox environment to execute tasks and report errors if any
- Created ideal responses for the LLM for RLHF
- Solving problems in different I1-I2-I3 taxonomy
- Solving hard PhD level problems
- Got 4.5/5 task rating
- Helping Google to improve Gemini by reviewing the LLM response by checking accuracy and instruction following

Sutherland Global Services

Data Scientist(Computer Vision/NLP)

Work Experience

May 2022 – November 2024

Working on Product CX306 which provide Data Analytics solution for Customer Data projects and help companies to Improve Customer Experience

Working on video and image processing for advertisement company using OCR , CNN , RetinaFace

- Implemented research paper <https://arxiv.org/html/2307.04570v3> utilising sagemaker
- Using Python and SQL to extract and manipulate data
- Working on OCR for getting text info from video and using transvNet for boundary shot detection for detecting changing shots in video , also using LLM to make OCR text informative and without grammatical error.
- Using Facer with FaRL for face , face feature extraction for optimising backbone instead of RetinaFace
- Using Sagemaker , S3 for storing and running OCR,using bounding boxes x-axis and y-axis to detect movement of text of sequence of frame.
- Training on LAION image dataset containing 10 TB of data , having 10 million images
- Using hugging face multiclass model for age prediction , building gender classification using CNN and VGG19 as backbone
- Testing retinaface face detection against aws rekognition and used image resizing ,fourior transform to preprocess image , Using histogramRB for image processing to get clusters of images for different illuminations, using FaRL as backbone for skintone detection for better facial embeddings.
- Using Tkinter for GUI of image and video for live recording of user.
- Working on NLP projects like CUNA,MARS , AFLAC
- Building Automation ML tools for production
- Building Chat System for customer support of AFLAC using Hugging face LLM
- Creating Synthetic data to train LLM for getting better Prompt result
- Using Transformer, diffuser packages, Similarity
- Creating defined functions like cosine_similarity , stop_words to automate process • Using Name Entity Recognition to perform he task and regex patterns
- Understanding the insurance related queries and giving model accuracy above 90%
- Finding baselines in chat data using regex
- Refining the baselines to get better prediction of model

- Using predefined similarity cosine function to predict the output Using stopword , stemming to preprocess the data
- Using hyperparameters like threshold , baselines , fixed_lenght to improve model accuracy
- Decreasing falsepositive(removing unnecessary baselines) and decreasing false negative by making changes in hyperparameters
- Using NLP packages like spacy,NER to get result
- Using OpenCV for object detection
- Web Scraping using Selenium , Webdriver, XPATH,action,Expected_Condition
- Using GCP cloud to deploy the model and kpis to the server
- Using GPT4 , GPT3, Falcon and trying to build the LLM model for chat system
- Working on Industrial level project which are based on Machine Learning and Deep Learning
- Predicting if a person get donation using Donor choose dataset: Predicting if a person get donation using Donorchoose dataset: Applying different models on donor choose data like GBDT, Naïve Bayes, Decision Tree and checking their accuracy in prediction Training, cross validating and testing the model TFIDF, W2V calculating AUC Hyperparameter tuning, grid search cv, Normaliser, count-vectoriser Performance matrix
- Time Series forecasting of logistics client to forecast sales using ARIMA
- Facebook Friend Recommendation: Predicting the relation of one person to another. XGBoost, RandomForestClassifier, Train, test, cross validate, AUC, ROC, Confusion Matrix, clf feature importances,Featurecreation .
- Product Recommendation System: Predicting the rating of product for new user and suggesting new productsbased on best rating SGD, SVD,matrix creation, Matrix Split, Matplotlib, Sklearn, MSE vs epoch.

Vinove Software

Work Experience

Dec 2018 – May 2022

Working on Python projects for Clients

- Building Python Script to auto generate excel report for monthly and yearly sales , revenue
- Adding data on daily bases and create error free report and deliver to client
- Data Collection using Selenium
- Feature Engineering using NLP techniques
- Implement word embedding using different techniques like TFIDF, Word2Vec, BERT
- Model Building and Performance Evaluation based on different metrics like Recall, Precision, ROC & AUC

Education

Murthal, Sonipat

DCRUST

Jun 2014 – Jun 2018

- B.Tech, 2019, CGPA: 7
- CBSE 12th, 2014, 82%
- CBSE 10th, 2012, CGPA: 9.6

PG Certification AI/ML

AppliedAi(Now Part of Scaler)

Jan 2021 – Jan 2022

- Working on Industrial level project which are based on Machine Learning and Deep Learning
- Predicting if a person get donation using Donor choose dataset: Predicting if a person get donation using Donor choose dataset: Applying different models on donor choose data like GBDT, Naïve Bayes, Decision Tree and checking their accuracy in prediction Training, cross validating and testing the model

TFIDF, W2V calculating AUC Hyperparameter tuning, grid search cv, Normaliser, count-vectoriser
Performance matrix

- Boston house price prediction: Creating random samples from the boston house price dataset. Training a random forest model with datapoints having the 13 features and then predicting for given datapoint using `clf.pred`
- A/B Testing , Hypothesis testing
- Working on Netflix dataset , understanding the schema of database , running SQL queries with optimisation

Technical Experience

- Experience in machine learning framework like TensorFlow, keras
- Neural Networks , Natural Language Processing and Algorithms
- Name Entity Recognition , text processing, Airflow
- OCR, RetinaFace, CNN, Object detection

Github - <https://github.com/varungarg239>

Certificate - <https://www.appliedaicourse.com/certificate/a499bb59d6>

Medium - <https://medium.com/@varunkmrgarg/time-series-in-healthcare-challenges-and-solutions503fdadd628>

Technical Skills

- Python, Pyspark , Numpy, Pandas, NLP, LLM, Deep Learning, GCP .
- SQL, Data Analytics, Statistics, Microsoft Excel, Data Visualisation
- Machine Learning, Classification, Regression Problem Solving
- OCR , Pre-Trained Model, Object Detection , Yolo, Sandbox

Achievements

Received Mathematics Silver Medal from FIITJEE

Hobbies

- Reading articles in AI/ML on medium , Kaggle , hugging Faces
- Playing Badminton , Lawn Tennis, Watching American Sitcom